

MONASH University

Optimal Sortition: A Framework Analysis

Daniel Loza
B.Bus., Monash University

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Abstract

Sortition, the random selection of individuals to form panels, has seen a sort of renaissance in recent years, as more and more institutions realise the value of direct community and stakeholder input when making decisions. These panels have proven to be massively beneficial in terms of improving the quality of decisions, increasing public trust and engagement, and creating more inclusive and representative decision-making processes. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective. This thesis aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, and what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

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Chapter 1

Introduction

1.1 Aim

This study aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

1.2 Background

We live in an age where decision-making and its impacts feel like they are only growing in importance and scale, yet our ability to impart influence seems to only be diminishing. Many feel disenfranchised and like power is being taken away from them, and that their voices are not being heard. This is a problem. Not only because it can lead to disengagement, apathy, and a lack of trust in institutions, but also because *not* listening directly to lived experiences and perspectives limits the effectiveness of the very decisions we *do* make.

According to Flanigan et al. (2021), sortition creates groups that are statistical "mini-publics" of society, reflecting demographics and perspectives more accurately than elected bodies. Which, given the polarised and over-dramatised nature of policy today, is a much-needed breath of fresh air.

Sortition is a method of selection that uses random sampling to create panels that are representative of the population. It has been used in various contexts, reaching as far back as ancient Athens, where it was how jurors were selected, and into modern times for citizens' assemblies. The idea is that by randomly selecting individuals from the population, we can create panels that are more representative

and inclusive than those selected through traditional methods. Sortition also reduces dominance by elites or highly engaged groups, enabling broader participation across social classes (Giuliani 2025), ensuring decisions actually serve the people they are meant to.

An increasing number of institutions (including the Parisian Citizens' Assembly, the Irish Citizens' Assembly, and the British Columbia Citizens' Assembly) are using sortition to create panels that are more representative of the population. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective.

One of the main challenges is ensuring that the selection process is transparent and fair, and that the panels are truly representative of the population in terms of demographics and perspectives. But this raises the questions of both a) what does it mean to be fair? and b) what does it mean to be representative?

Additionally, though some (such as LexiMin, from Flanigan et al. (2021)) have already been developed, there is a need for better tools, methods and frameworks to evaluate the representativeness of these panels and to ensure that they are robust in achieving their goals.

Chapter 2

Literature Review

2.1 Representation and Sortition

2.1.1 What does it mean to be representative?

The main aim of any and all selection methods is to create a panel that is fair and representative of the population. But what does this actually mean?

Fairness refers to the idea that the selection process should be transparent, unbiased, and inclusive. It should give everyone an equal chance of being selected, regardless of their background or characteristics. Which is easier said than done.

Alongside this, representativeness refers to the idea that the selected panel should accurately reflect the demographics and perspectives of the population. This means that the panel should be diverse and inclusive, and should contain individuals from a wide range of relevant backgrounds.

Approaching it systematically, the sortition process can be thought of in four stages: - Defining the population and traits of interest - Sampling from the population (through survey or other outreach) - Panel selection (through algorithmic means or other method) - Panel evaluation, confirmation and engagement

Each of these steps presents an opportunity for bias to be introduced, and for the panel to become less representative of the population. Which is why it is crucial to evaluate the level of representativeness (or *potential* representativeness) at each stage, and to understand how different factors may affect it.

For example, if the population of interest and investigated traits are poorly defined, then the selection process may not be able to capture the diversity and perspectives of the desired group, leading to a less representative panel. This presents a challenge in itself, given most population descriptions (as

previously discussed) are often based on disjoint demographic characteristics and do not provide a way to capture intersectionality, which is a key aspect of representativeness. This may be beyond the scope of this thesis, as it is already a foray into broader data collection and storage methods, but it is important to keep in mind that representation goes beyond the margin.

Similarly, if certain groups are less likely to respond to the survey, then they may be underrepresented in the pool of potential panelists, which can lead to a less representative panel from the outset. Illustratively–if there were to be no female respondents for a certain survey in an area comprised of 60% women, then the representativeness assumption (and goal) would be majorly violated before an attempt at putting a panel together has even been made. Situations like these raise the question of how we can account for this non-response bias, and whether we can use visualisation tools to show the discrepancy between the pool and the population (which leads back to part of this thesis' aim).

2.2 Panel Selection

2.2.1 Problem Definition

Suppose there is a pool of individuals numbered 1 to n.

We want to choose a panel of size k.

Let K be the set of all feasible panels. A feasible panel is any group of k individuals that satisfies the required constraints (which generally take the form of quotas).

2.2.2 LexiMin Approach (Flanigan et al. 2021)

LexiMin does not choose one panel immediately. Instead, it chooses a probability distribution over all feasible panels.

Each feasible panel is assigned a probability.

These probabilities must be non-negative and must add up to 1.

For each individual, we then calculate their overall probability of being selected by adding up the probabilities of all feasible panels that include that individual.

Next, we sort the individual selection probabilities from smallest to largest.

The LexiMin rule works by first making the smallest selection probability as large as possible. Subject to that, it makes the second-smallest as large as possible. It then does the same for the third-smallest, and continues in this way.

2.2.3 Formal Representation

Let p_S be the probability assigned to panel S.

These probabilities satisfy:

$$p_S \geq 0$$

and

$$\sum_{S \in K} p_S = 1$$

The selection probability for individual i is:

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S$$

This gives the vector of individual selection probabilities:

$$\pi(p) = (\pi_1(p), \pi_2(p), ..., \pi_n(p))$$

Sort these from smallest to largest:

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq ... \leq \pi_{(n)}(p)$$

LexiMin chooses p so that this sorted vector is maximised in order.

2.2.4 Algorithm

1. List all feasible panels in K.
2. Consider all possible probability distributions over those panels.
3. For each distribution, compute:

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S$$

4. Sort the selection probabilities:

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq \ldots \leq \pi_{(n)}(p)$$

5. Choose the distribution that maximises the smallest selection probability.
6. If more than one distribution still remains, choose the one that maximises the second-smallest selection probability.
7. Continue in this way until the distribution is determined.
8. Sample a final panel according to that distribution.

2.2.5 Logic

LexiMin is a fairness rule. It gives priority to the individuals with the lowest chance of being selected and improves their chances first. Then it improves the next-lowest probabilities, and so on.

This makes the selection probabilities as equal as possible while still respecting the panel constraints.

2.2.6 Discussion

This approach to the problem of panel selection is elegant and has some nice properties. It ensures that the selection probabilities are as equal as possible, which can help to create a more representative panel. However, it does not give a way to introduce specific joint trait requirements for subsets of the population, which is something that may be desirable in certain situations. For example, if there is a need to ensure that a certain combination of traits is represented in the panel, then this approach may not be able to accommodate that requirement without reintroducing representative burden.

2.3 Visualisation Methods

2.3.1 Parallel coordinate plots (VanderPlas et al. 2023)

- Can help to show the distribution of traits across the selected panel and how it compares to the overall population.
- Can also show the trade-offs between different traits and how they are balanced in the selection process
- Can be used to identify any segments of the population that are underrepresented in the selected panel, and to highlight any potential biases in the selection process.

2.3.2 Matrix-based set representation (UpSetR) (Conway, Lex & Gehlenborg 2017)

- Can be used to show how different traits connect with each other and how they are distributed across the distribution and/or selected panel.

- Can help to identify any clusters of traits that are overrepresented or underrepresented in the selected panel, and to show how different traits interact with each other in the selection process.
- Can highlight specific combinations of traits that are of interest, and show how they are represented in the selected panel compared to the overall population. (Though there is no way to gauge the prevalence of these combinations in the population, beyond anecdotal observations)

2.3.3 Query-based highlighting

- Makes the visualisation interactive, allowing users to explore the data and identify specific patterns or trends in the selection process.
- Can be used to highlight specific traits or combinations of traits that are of interest, and to show how they are represented in the selected panel compared to the overall population.
- Also creates opportunity to ensure specific combinations of traits are represented in the selected panel, without reintroducing representative burden.

2.3.4 Attribute distribution analysis

- Showcase how the distribution of traits in the selected panel compares to the overall population, and to identify any discrepancies or biases in the selection process.

2.4 Working questions

2.4.1 Sortition

- For the new quota tool, would it just be a matter of making a concatenated trait quota, evaluating whether the resulting panel is still feasible and meets the requirements of the original quotas, and then tested to see whether it majorly impacted selection likelihood, vs if it were not to have been used?

2.4.2 Visualisation

- What do we actually want to show with these visualisations? What are the key insights we want to communicate?
- Are we looking at likelihood of selection across traits/through stages? Or are we looking at the distribution of traits in the final panel?

Chapter 3

Research Question

What does it mean for a panel to be “good” at being representative and are we achieving this? - Are current methods and algorithms, like LexiMin, performing the best they could be? - How do unobserved/unfiltered traits affect response rates, and therefore panel composition and representativeness? - How can survey non-response and other sampling-level issues bias representativeness?

nDF suggests 10x response to panel count; not generally feasibly, more often get 3x. How much of a difference does this make to burden and selection likelihood?

Chapter 4

Research Plan

Plan

Chapter 5

Preliminary Results

This chapter contains a summary of the context in which your research is set.

Bibliography

Conway, JR, A Lex & N Gehlenborg (2017). UpSetR: An R Package for the Visualization of Intersecting Sets and Their Properties. *Bioinformatics* **33**(18). Ed. by J Hancock, 2938–2940. (Visited on 04/09/2026).

Flanigan, B, P Gölz, A Gupta, B Hennig & AD Procaccia (2021). Fair algorithms for selecting citizens' assemblies. *Nature* **596**, 548–552.

Giuliani, A (2025). Sortition in Politics: From History to Contemporary Democracy. *Parliaments, Estates and Representation* **45**(3), 317–335.

VanderPlas, S, Y Ge, A Unwin & H Hofmann (2023). Penguins Go Parallel: A Grammar of Graphics Framework for Generalized Parallel Coordinate Plots. *Journal of Computational and Graphical Statistics* **32**(4), 1572–1587. (Visited on 04/09/2026).